

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.8 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 One root of the equation $t^2 - (7 \sin^{-1} x)t + (12\pi \sin^{-1} x + f(x)) = 0$ is six times of other root. If M and

m are maximum and minimum value of $f(x)$ then the value of $\frac{2(M+m)}{\pi^2}$ equals

- (A) 15 (B) 11 (C) 8 (D) 6

Q.2 If $f(x) = ax^2 + bx + c$ is such that $f^2(-1) + f^2(0) + f^2(1) + 30 = 4f(0) + 10f(1) + 2f(-1)$, then

- (A) range of $\sin^{-1}(f(x))$ is $\left\{\frac{\pi}{2}\right\}$ (B) range of $\cos^{-1}(f(x))$ is $\{\pi\}$
(C) range of $\tan^{-1}(f(x))$ is $\left[-\frac{\pi}{2}, \frac{\pi}{4}\right]$ (D) range of $\cot^{-1}(f(x))$ is $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

Q.3 Fundamental period of the function

$$f(x) = \cos^{-1}(\cos \pi x) + |\sin(\pi x)| + \operatorname{sgn}(x^2 - 10x + 30) + 2013 \sin \pi [x]$$

[Note: $[x]$ denotes greatest integer less than or equal to x and $\operatorname{sgn} x$ denotes signum function of x .]

- (A) π (B) 1 (C) 2π (D) 2

Q.4 Let $f(x)$ satisfies $\frac{f^2(x)}{f(1-x)} = x^3$, where $f(1-x) \neq 0$, then $f\left(\frac{1}{2}\right)$ equals

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{64}$

Q.5 Let $f(x) = ([\lambda]^3 - 9[\lambda]^2 + 26[\lambda] - 24)x^5 - (24\{\lambda\}^3 - 26\{\lambda\}^2 + 9\{\lambda\} - 1)x^3 + x^2 + \cos^6 \pi x - 2013$ is an even function then sum of all possible values of λ equals

[Note: $[x]$ and $\{x\}$ denotes greatest integer less than or equal to x and fractional part of x respectively.]

- (A) $\frac{131}{4}$ (B) $\frac{151}{4}$ (C) $\frac{121}{4}$ (D) $\frac{101}{4}$

Q.6 Consider $f(x) = |x||x-1||x-2||x-3| \forall x \in [1, 2]$ and $g(x) = |x| + |x-1| + |x-2| + |x-3| \forall x \in (-\infty, \infty)$. If $\max. f(x) = \lambda \min g(x)$ at $x = x_0$ then $(\lambda + x_0)$ is equal to

- (A) $\frac{15}{64}$ (B) $\frac{9}{64}$ (C) $\frac{105}{64}$ (D) $\frac{3}{2}$

- Q.7 If $\alpha = \sin\left(\cot^{-1}\left(\tan\left(\cos^{-1}\frac{2}{3}\right)\right)\right)$ and $\beta = \sin\left(\operatorname{cosec}^{-1}\left(\cot\left(\tan^{-1}\frac{1}{3}\right)\right)\right)$ are the roots of the quadratic equation $ax^2 + bx + c = 0$ where a, b, c are integers and c is prime then the value of $(a + b + c)$ equals
 (A) 1 (B) 2 (C) 3 (D) 9
- Q.8 If f is a continuous function in $[1, 2]$ such that $|f(1) + 3| < |f(1)| + 3$ and $|f(2) + 10| = |f(2)| + 10$, $(f(2) \neq 0)$ then the equation $f(x) = 0$ in $(1, 2)$ has
 (A) at least one real root (B) no real root
 (C) exactly two real roots (D) exactly one real root

[PARAGRAPH TYPE]

Q.9 to Q.14 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 9 to 11

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that, $f(x) = x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$ be a polynomial function. If $f(1) = f(2) + 2$ and $f(2) = f(3) + 3$, then answer the following problems.

- Q.9 If $f(3) = f(4) + k$, then k is equal to
 (A) -2 (B) -5 (C) 2 (D) 5
- Q.10 Value of $(a + b)$ is
 (A) 2 (B) 3 (C) 4 (D) 5
- Q.11 $f(x)$ is
 (A) one one and onto function (B) one one and into function
 (C) many one and onto function (D) many one and into function

Paragraph for question nos. 12 to 14

For the function $f(x) = 2x^2 - 2(\tan^{-1} \alpha - \cot^{-1} \alpha)x - (\tan^{-1} \alpha)(\cot^{-1} \alpha) \quad \forall \alpha \in [-1, 1]$ vertex of the curve $y = f(x)$ is at $P \equiv (h, k)$

- Q.12 Which is always correct?
 (A) $h > 0$ (B) $h \geq 0$ (C) $h < 0$ (D) $h \leq 0$
- Q.13 Equation $f(x) = 0$ has always
 (A) two real and distinct roots (B) two equal roots
 (C) two imaginary roots (D) nothing can be said
- Q.14 If $\frac{16k}{\pi^2} = \lambda$, then λ can take how many integral values
 (A) 5 (B) 4 (C) 3 (D) 2

[MULTIPLE CORRECT CHOICE TYPE]

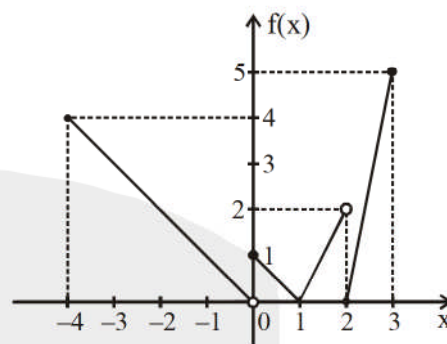
Q.15 to Q.17 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

Q.15 If $f^2(x) \cdot f\left(\frac{1-x}{1+x}\right) = x^3 \quad \forall x \in \mathbb{R} - \{-1, 1\}$ and $f(x) \neq 0$, then which of the following is correct.

- (A) $f(2) \cdot f(-2) = 16$ (B) $f(2)f(-2) = 8$ (C) $f(2) = -12$ (D) $f(3) = -18$

Q.16 The graph of a function $y = f(x)$ is shown in the figure Identify which of the following statement is/are correct?

- (A) Domain of $f(f(x))$ is $[-4, 3]$.
 (B) Domain of $f(f(x))$ is $\left[-3, \frac{13}{5}\right]$.
 (C) The equation of $f(f(x)) = 1$ has exactly 7 solutions.
 (D) The equation of $f(f(x)) = 1$ has exactly 8 solutions.



Q.17 $\tan^{-1} \frac{1}{8} + \tan^{-1} \frac{1}{18} + \tan^{-1} \frac{1}{32} + \tan^{-1} \frac{1}{50} + \dots \infty$ equals

- (A) $\frac{\pi}{2} - \tan^{-1} 3$ (B) $\frac{1}{2} \tan^{-1} \frac{3}{4}$ (C) $\tan^{-1} \frac{1}{3}$ (D) $\frac{1}{2} \cos^{-1} \frac{4}{5}$

[MATRIX TYPE]

Q.18 has four statements (A, B, C, D) given in **Column-I** and five statements (P, Q, R, S, T) given in **Column-II**. Any given statement in **Column-I** can have correct matching with one or more statement(s) given in **Column-II**.

Q.18 Consider the locus of a moving point $P(x, y)$ satisfying $25 \left((x-2)^2 + (y-3)^2 \right) = C^2 (3x - 4y + 7)^2$ where C is a function of λ .

[Note: $[k]$ & $\{k\}$ denotes the greatest integer less than or equal to k and fractional part of k respectively.] Match the items in Column-I with those in Column-II.

Column-I	Column-II
(A) If C is $[\sin^{-1} \lambda]$, then P moves on	(P) Parabola
(B) If C is $[\cot^{-1} \lambda]$, then P moves on	(Q) Ellipse
(C) If C is $\frac{1}{[\cos \lambda]}$, then P moves on	(R) Hyperbola
(D) If C is $\frac{1}{[\lambda^2]}$, then P moves on	(S) Straight Line
	(T) Point

[INTEGER TYPE]

Q.19 to Q.21 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

Q.19 Consider, $f(x) = \begin{cases} \frac{3}{2} \sin^{-1}\left(\frac{2x}{1+x^2}\right) + 3 \tan^{-1} x, & x > 1 \\ \frac{3\pi}{2} x, & -1 \leq x \leq 1 \\ \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) + 2 \tan^{-1} x - \frac{3\pi}{2}, & x < -1 \end{cases}$

If area enclosed by the graph of $f(x)$ and the x-axis from $x = -2$ to $x = 2$ is $\left(\frac{p}{q}\right)\pi$, where $p, q \in \mathbb{N}$, then find the least value of $|p - q|$.

Q.20 Find the sum of all the roots of the equation $[e^{-x}] = \{x\}$ when $x \in [-100, 100]$.
[Note: $[x]$ & $\{x\}$ denotes greatest integer function & fractional part of function of x .]

Q.21 If $f: [4, a] \rightarrow A$ is a bijective function and defined by $f(\sqrt{x-1} + \sqrt{17-x}) = \sqrt{20 + 2\sqrt{64 - (x-9)^2}}$, then find the value of $[f^{-1}(5)]$.

[Note : $[k]$ denotes greatest integer less than or equal to k .]

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ANSWER KEY

Q.1	D	Q.2	A	Q.3	D	Q.4	C	Q.5	C
Q.6	C	Q.7	B	Q.8	A	Q.9	A	Q.10	C
Q.11	C	Q.12	D	Q.13	A	Q.14	A	Q.15	ACD
Q.16	BC	Q.17	ABCD	Q.18	(A) P, T, R ; (B) P, R, T ; (C) P ; (D) P, Q				
Q.19	7	Q.20	5050	Q.21	4				

